MIL-HDBK-271-105
2 DECEMBER 1985

MILITARY HANDBOOK

AIRCRAFT FASTENERS USAGE AND INSTALLATION FOR MIL-S-85069 FASTENER SLEEVES



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DEPARTMENT OF DEFENSE
WASHINGTON 25, DC 20361

MIL-HDBK-271-105

Aircraft Fasteners Usage and Installation Methods for MIL-S-85069 Fastener Sleeves

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.

2. The publication was approved on 1 December 1985 for printing and inclusion in the military standardization handbook series.

3. This handbook provides usage and installation methods for fastener sleeves. It will provide valuable information and guidance to aircraft mechanics at Navy, Army and Air Force Rework Facilities.

   This handbook is not intended to be referenced in procurement specifications except for information purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to include the latest information on usage and installation of fastener sleeves. Mechanics should review this handbook periodically. Users of this handbook are encouraged to report any errors or recommendations to: Commanding Officer, Naval Air Engineering Center, Code 93, SESD, Lakehurst, NJ 08733.

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Scope. This handbook covers the usage and installation of MIL-S-85069 fastener sleeves.

Description and purpose. These fastener sleeves are thin wall, tubular elements with a flared end. These sleeves were developed to meet several specific needs such as bushing oversize holes, protection from corrosion, and to aid sealing.

The primary use of fastener sleeves is to repair enlarged holes. The sleeves come in standard fastener diameters with standard fastener heads. The sleeves are generally one inch long and are adjusted to the length of fastener being used. Sleeves may be bonded into areas where holes need protection such as vent holes, drain holes or through composite structure.

Fastener sleeves are manufactured from A286 stainless steel. There are four head styles, three flush and one protruding (see Tables IA and IB). The sleeves have grooves on the shank spaced 1/16 of an inch apart. One function of these grooves is to trap sealant between the sleeve and the wall of the hole resulting in better seals. The second function is to aid in adjusting the sleeve to length. MIL-S-85069 and associated specification sheets cover the fastener sleeves, as shown in Table II.

For an explanation of notes appearing on tables and figures, see page 21.

Hole preparation. Drill or ream the hole for the fastener and sleeve in accordance with Tables III through X, adapted from the applicable specification sheets. If flush head fasteners are to be installed, the countersink in the material will have to be deepened 0.005 inches for 1/64 and 0.010 inches for 1/32 oversize to accommodate the countersunk portion of the sleeve. The installed sleeve and fastener shall meet the applicable aerodynamics smoothness requirements on exterior surfaces. Except where a smoother finish is specified on the engineering drawing or in the fastener installation document, the surface finish of the hole shall not exceed 125 RHR as defined in ANSI-B-46.1.

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The sleeve shall correspond to the fastener configuration (flush or protruding head), fastener application (tension or shear), and fastener diameter. See the appropriate table (III through X). Each type of sleeve is stocked in only one length. Diameters #10 through 3/8 inch have a one-inch grip length. All sleeves have external circumferential grooves spaced 1/16 inch apart over the straight portion of their grip length.

Sleeves shall be prepared for installation by breaking or grinding them off at a groove so that the grip length will be the same as the fastener grip length. If the grip length of the fastener is longer than the maximum length of the sleeve, an additional cylindrical section broken from a second sleeve may be used to make up the required fastener grip length. Do not use a section of sleeve less than 1/4 of an inch long. Break off tools listed in Tables IA and IB must be used.

Adjusting to length. Using step plate and breakoff handle (see Tables IA and IB), break sleeve to length per figure 1.

Alternate method of adjusting sleeve to length in structure. To adjust sleeve to length in a structure, be sure the structure sheets are firmly held together. Insert the sleeve into the hole until the head is fully seated. Examine the rear protrusion of the sleeve. If a groove is barely visible on the sleeve next to the sheet, use the breakoff handle to break off the protruding section of sleeve. If the back surface of the sheet falls in between grooves, place a standard washer, 1/32 inch thick, on the sleeve and push securely against the back sheet. Then using the breakoff handle, break off the portion of the sleeve standing above the washer. This will always leave the sleeve protruding slightly from the back sheet. The washer may be left on or may be removed and a nut or collar with the proper counterbore used on the bolt.

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Procedure for grinding sleeve to proper length. See Figure 2.

Installation. Fastener sleeves must be installed with reasonable care to insure that they function properly in inhibiting corrosion, sealing and in serving to restore the structural integrity of joints in modification, repair or in the original manufacture of aerospace or military hardware.

Fasteners and sleeves may be installed wet with sealing compounds as required. A thin coat of compound shall be applied to the exterior of the sleeve and the sleeve inserted into the hole. If there is an interference fit between the sleeve and the hole, inserting tool JK6543-XX shall be used to install the sleeve into the hole. The same material shall be applied to the shank of the fastener and the fastener inserted or driven into the sleeve. Do not allow the compound to cure before completing the installation. (See figure 3.)

Flushing back side when necessary. Normally the sleeve does not need to be flush on the back side. The sleeve protrusion will fit into the same counterbore in the attaching nut or collar as an oversize bolt. A washer may be used when necessary.

When the sleeve must be flush on the back side it may be trimmed by using a counterbore or countersink in a cage. Adjust the cage to cut flush. A pilot on the cutting tool to fit the sleeve hole must be used. Cut slowly.

To keep the sleeve from turning, it is helpful to have it in an interference fit, bonding is suggested.

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Sleeves, fasteners, design and usage limitations. The fastener sleeves shall replace oversize shank fasteners for repair and maintenance. The sleeves shall be used with standard diameter straight shank fasteners. They shall be used under the same restrictions as oversize fasteners and in compliance with the specific user's repair procedures and restrictions.

The fastener or attaching element (nut or collar) shall contain a counterbore to accept the protrusion of the sleeve; or a washer shall be used; or the sleeve shall be flush on the backside of the stack up.

The fastener sleeves shall not be used with tapered shank fasteners or blind fasteners.

Interference (hole to sleeve) shall be not greater than .003" where the material thickness is less than 1/2". Interference shall be not greater than .0015" where material is greater than 1/2".

Appropriate engineering approval shall be required in the following applications:

a. In fluid containers having no self-sealing feature (e.g., fuel, oil or water tanks).

b. Fatigue and fracture critical locations.

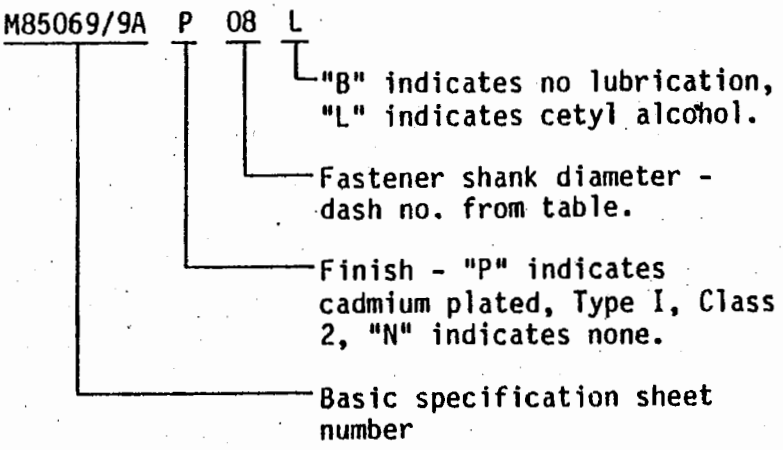
c. Where the edge distance from the center of the hole is less than 2D.

d. Where the number of enlarged holes being repaired is greater than 10% of the total number of holes in a pattern.

e. Where more than three adjacent holes are enlarged.

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Part number example.



The sleeves discussed herein are manufactured under United States Patent Number 3835615 and 3835525. Both patents expire on 9/1/91. The Government does not have a royalty-free license.

Tooling procurement. The tools referenced in this handbook are available from:

J. O. King, Incorporated
P. O. Box 93826
711 Trabert Avenue, N.W.
Atlanta, GA 30318

Custodians:
Navy - AS
Air Force - 11
Army - AV

Preparing Activity:
Navy - AS
(Project No. 53GP-0149-05)

Review Activities:
DLA - IS
Air Force - 99
Army - AR

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FIRST INCREMENT OVERSIZE - 1/64

FOR USE WITH GROOVED SLEEVES

MIL-S-85069 Configuration	For Fasteners With These Head Styles	Sleeve Vendor Part Number (5)	Tooling	Description	Tool Part Number (5)
MIL-S-85069/2	Protruding Head Fasteners Shear And Tension	JK5517CXX		Step Plate	JK6555-XX
				Step Plate For 05,55 Dia. Only	JK6557-05 JK6557-55
MIL-S-85069/3	Reduced Shear Flush Head (NAS1097)	JK5512CXX		Break Off Handle	JK6548-XX
MIL-S-85069/4	Shear Flush Head (MS20426)	JK5514CXX		Inserting Tool Interference Only	JK6542-XX
MIL-S-85069/5	Tension Flush Head (AN509)	JK5516CXX		Removing Tool	JK6526-XX

TABLE IA. Fastener sleeve styles.

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SECOND INCREMENT OVERSIZE 1/32 FOR USE WITH GROOVED SLEEVES					
MIL-S-85069 Configuration	For Fasteners With These Head Styles:	Sleeve Vendor Part Number	Tooling	Description	Tool Part Number (5)
MIL-S-85069/6	Protruding Head Fasteners Shear And Tension	JK5577CXX		Step Plate	JK6556-XX
MIL-S-85069/7	Reduced Shear Flush Head (NAS1097)	JK5572CXX		Break Off Handle	JK6546-XX
MIL-S-85069/8	Shear Flush Head (MS20426)	JK5574CXX		Inserting Tool	JK6543-XX
MIL-S-85069/9	Tension Flush Head (AN509)	JK5576CXX		Removing Tool	JK6524-XX

TABLE IB. Fastener sleeve styles.

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SPECIFICATION SHEETS	
MIL-S-85069/1A	Sleeves, Fastener, Grooved, Design and Usage Limitations for
1/64 OVERSIZE	
MIL-S-85069/2A	Sleeves, Fastener, Grooved, Protruding Head, Shear and Tension, 1/64 Oversize
MIL-S-85069/3A	Sleeves, Fastener, Grooved, 100° Flush Reduced Head, Shear, 1/64 Oversize
MIL-S-85069/4A	Sleeves, Fastener, Grooved, 100° Flush Head, 1/64 Oversize
MIL-S-85069/5A	Sleeves, Fastener, Grooved, 100° Flush Full Head, Tension, 1/64 Oversize
1/32 OVERSIZE	
MIL-S-85069/6A	Sleeves, Fastener, Grooved, Protruding Head, Shear and Tension, 1/32 Oversize
MIL-S-85069/7A	Sleeves, Fastener, Grooved, 100° Flush Reduced Head, Shear, 1/32 Oversize
MIL-S-85069/8A	Sleeves, Fastener, Grooved, 100° Flush Head, 1/32 Oversize
MIL-S-85069/9A	Sleeves, Fastener, Grooved, 100° Flush Full Head, Tension, 1/32 Oversize

TABLE II. Military specification sheets.

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GROOVED

DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	L ±.010	RECOM. HOLE PREP. 3	
							DRILL NO.	DRILL DIA.
05	5/32	.312	.1565 .1550	.1730 .1720	.125	.750	17	.1730
55	#8	.325	.1635 .1620	.1800 .1790	.125	.750	15	.1800
06	#10	.380	.1895 .1880	.2055 .2045	.125	1.00	5	.2055
08	1/4	.500	.2495 .2480	.2656 .2646	.188	1.00	17/64	.2656
10	5/16	.620	.3120 .3105	.3281 .3271	.250	1.00	21/64	.3281
12	3/8	.750	.3745 .3730	.3906 .3896	.312	1.00	25/64	.3906

TABLE III. Protruding head, shear and tension 1/64 oversize. (MIL-S-85069/2A)

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DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	H_F HEAD HEIGHT FAST. (REF)	H_S HEAD HEIGHT SLEEVE (REF)	L ±.010	RECOM. HOLE PREP. [3]	
									DRILL NO.	DRILL DIA.
05	5/32	.252	.1565 .1550	.1730 .1720	.125	.040	.045	.750	17	.1730
55	#8	.260	.1635 .1620	.1800 .1790	.125	.041	.046	.750	15	.1800
06	#10	.300	.1895 .1880	.2055 .2045	.125	.047	.052	1.00	5	.2055
08	1/4	.393	.2495 .2480	.2656 .2646	.188	.061	.066	1.00	17/64	.2656
10	5/16	.472	.3120 .3105	.3281 .3271	.250	.068	.073	1.00	21/64	.3281
12	3/8	.558	.3745 .3730	.3906 .3896	.312	.078	.083	1.00	25/64	.3906

TABLE IV. 100° flush reduced head, shear 1/64 oversize. (MIL-S-85069/3A)

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GROOVED

DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	H_F HEAD HEIGHT FAST. (REF)	H_S HEAD HEIGHT SLEEVE (REF)	L ±.010	RECOM. HOLE PREP. 3	
									DRILL NO.	DRILL DIA.
05	5/32	.282	.1565 .1550	.1730 .1720	.125	.053	.058	.750	17	.1730
55	#8	.294	.1635 .1620	.1800 .1790	.125	.055	.060	.750	15	.1800
06	#10	.347	.1895 .1880	.2055 .2045	.125	.071	.076	1.00	5	.2055
08	1/4	.470	.2495 .2480	.2656 .2646	.188	.098	.103	1.00	17/64	.2656
10	5/16	.558	.3120 .3105	.3281 .3271	.250	.109	.114	1.00	21/64	.3281
12	3/8	.688	.3745 .3730	.3906 .3896	.312	.137	.142	1.00	25/64	.3906

TABLE V. 100° flush head, shear (426) 1/64 oversize. (MIL-S-85069/4A)

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DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	H_F HEAD HEIGHT FAST. (REF)	H_S HEAD HEIGHT SLEEVE (REF)	L ±.010	RECOM. HOLE PREP. 3	
									DRILL NO.	DRILL DIA.
05	5/32	.312	.1565 .1550	.1730 .1720	.125	.065	.070	.750	17	.1730
55	#8	.328	.1635 .1620	.1800 .1790	.125	.070	.075	.750	15	.1800
06	#10	.380	.1895 .1880	.2055 .2045	.125	.080	.085	1.00	5	.2055
08	1/4	.505	.2495 .2480	.2656 .2646	.188	.108	.113	1.00	17/64	.2656
10	5/16	.631	.3120 .3105	.3281 .3271	.250	.135	.140	1.00	21/64	.3281
12	3/8	.758	.3745 .3730	.3906 .3896	.312	.162	.167	1.00	25/64	.3906

TABLE VI. 100° flush full head, tension 1/64 oversize. (MIL-S-85069/5A)

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DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	L ±.010	RECOM. HOLE PREP. 3	
							DRILL NO.	DRILL DIA.
06	#10	.380	.1895 .1880	.2210 .2200	.125	1.00	2	.2210
08	1/4	.500	.2495 .2480	.2812 .2802	.188	1.00	9/32	.2812
10	5/16	.620	.3120 .3105	.3437 .3427	.250	1.00	11/32	.3437
12	3/8	.750	.3745 .3730	.4062 .4052	.312	1.00	13/32	.4062

TABLE VII. Protruding head, shear and tension 1/32 oversize. (MIL-S-85069/6A)

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DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	H_F HEAD HEIGHT FAST. (REF)	H_S HEAD HEIGHT SLEEVE (REF)	L ±.010	RECOM. HOLE PREP. 3	
									DRILL NO.	DRILL DIA.
06	#10	.300	.1895 .1880	.2210 .2200	.125	.047	.057	1.00	2	.2210
08	1/4	.393	.2495 .2480	.2812 .2802	.188	.061	.071	1.00	9/32	.2812
10	5/16	.472	.3120 .3105	.3437 .3427	.250	.068	.078	1.00	11/32	.3437
12	3/8	.558	.3745 .3730	.4062 .4052	.312	.078	.088	1.00	13/32	.4062

TABLE VIII. 100° flush reduced head, shear 1/32 oversize. (MIL-S-85069/7A)

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DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	H_F HEAD HEIGHT FAST. (REF)	H_S HEAD HEIGHT SLEEVE (REF)	L ±.010	RECOM. HOLE PREP. 3	
									DRILL NO.	DRILL DIA.
06	3/16	.347	.1895 .1880	.2210 .2200	.125	.071	.081	1.00	2	.2210
08	1/4	.470	.2495 .2480	.2812 .2802	.188	.098	.108	1.00	9/32	.2812
10	5/16	.558	.3120 .3105	.3437 .3427	.250	.109	.119	1.00	11/32	.3437
12	3/8	.688	.3745 .3730	.4062 .4052	.312	.137	.147	1.00	13/32	.4062

TABLE IX. 100° flush head, shear (426) 1/32 oversize. (MIL-S-85069/8A)

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DASH NO.	FOR FAST. SIZE (REF)	A DIA. ±.005	D_I DIA.	D_O DIA.	G	H_F HEAD HEIGHT FAST. (REF)	H_S HEAD HEIGHT FAST. (REF)	L ±.010	RECOM. HOLE PREP. ③	
									DRILL NO.	DRILL DIA.
06	#10	.380	.1895 .1880	.2210 .2200	.125	.080	.090	1.00	2	.2210
08	1/4	.505	.2495 .2480	.2812 .2802	.188	.108	.118	1.00	9/32	.2812
10	5/16	.631	.3120 .3105	.3437 .3427	.250	.135	.145	1.00	11/32	.3437
12	3/8	.758	.3745 .3730	.4062 .4052	.312	.162	.172	1.00	13/32	.4062

TABLE X. 100° flush full head, tension 1/32 oversize. (MIL-S-85069/9A)

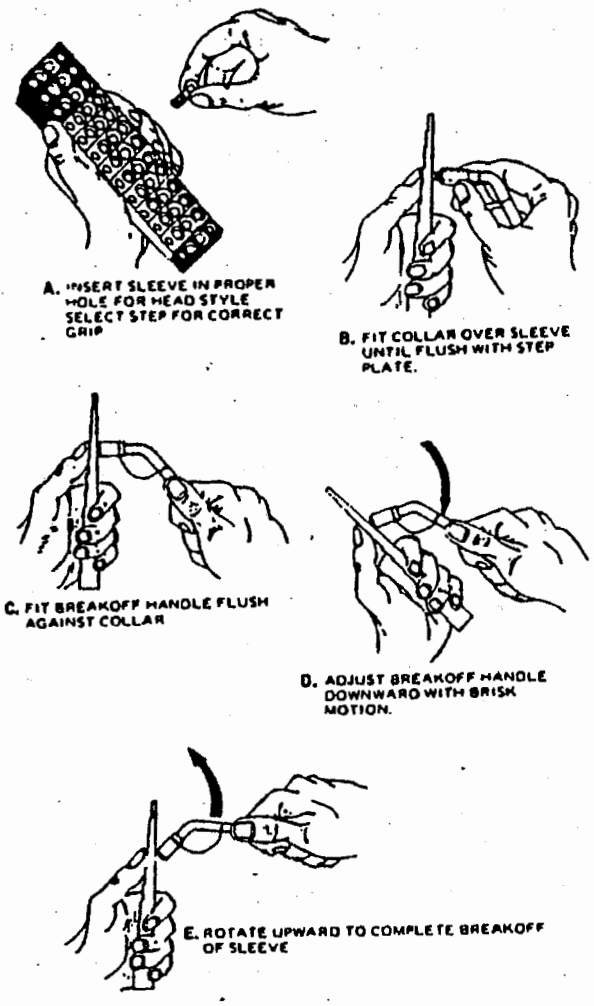
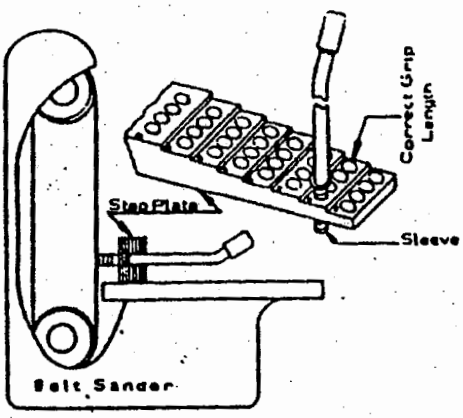


FIGURE 1. Adjusting grooved sleeves to length.

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NOTES
1. May also be ground on disc sander.
2. Deburr inside and out.
3. Do not grind the bottom of the step plate.

FIGURE 2. Grinding sleeves to proper length.

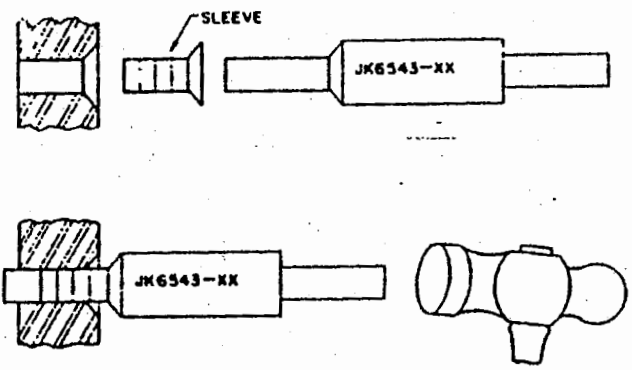
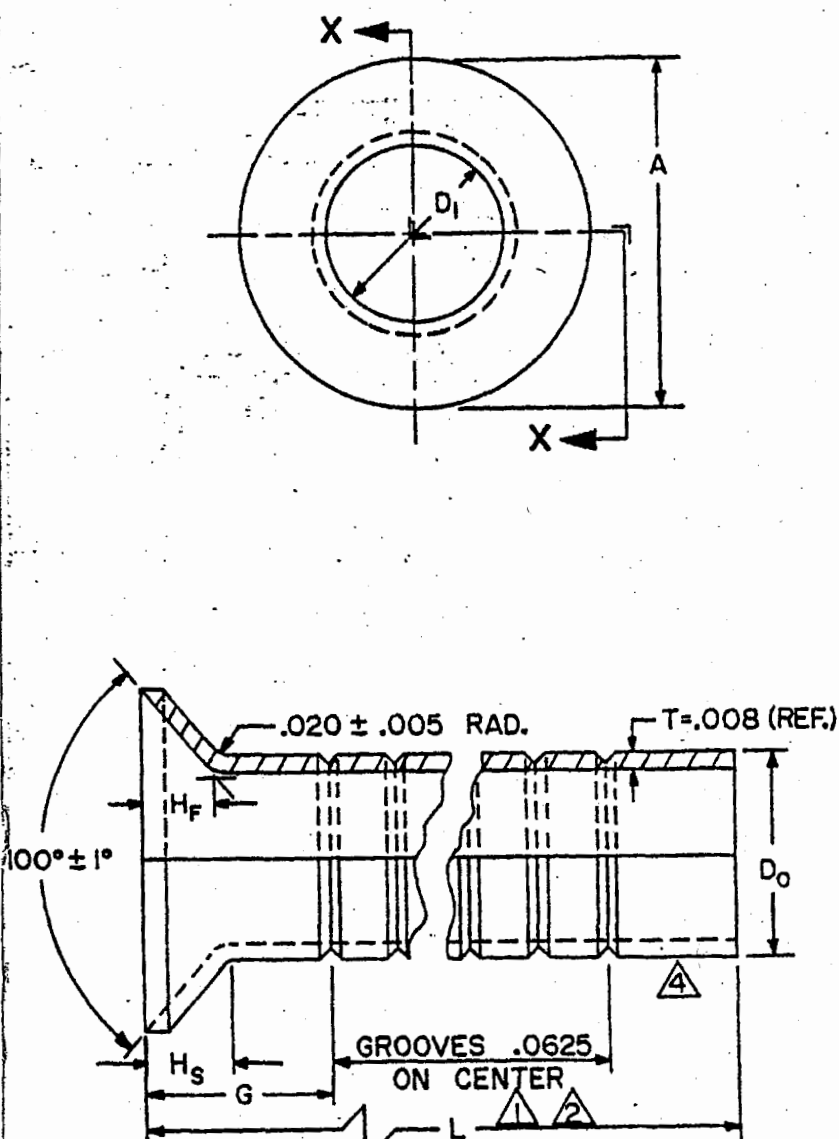


FIGURE 3. Installing fasteners and sleeves.

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SECTION X-X

FIGURE 4. Flush head, grooved.

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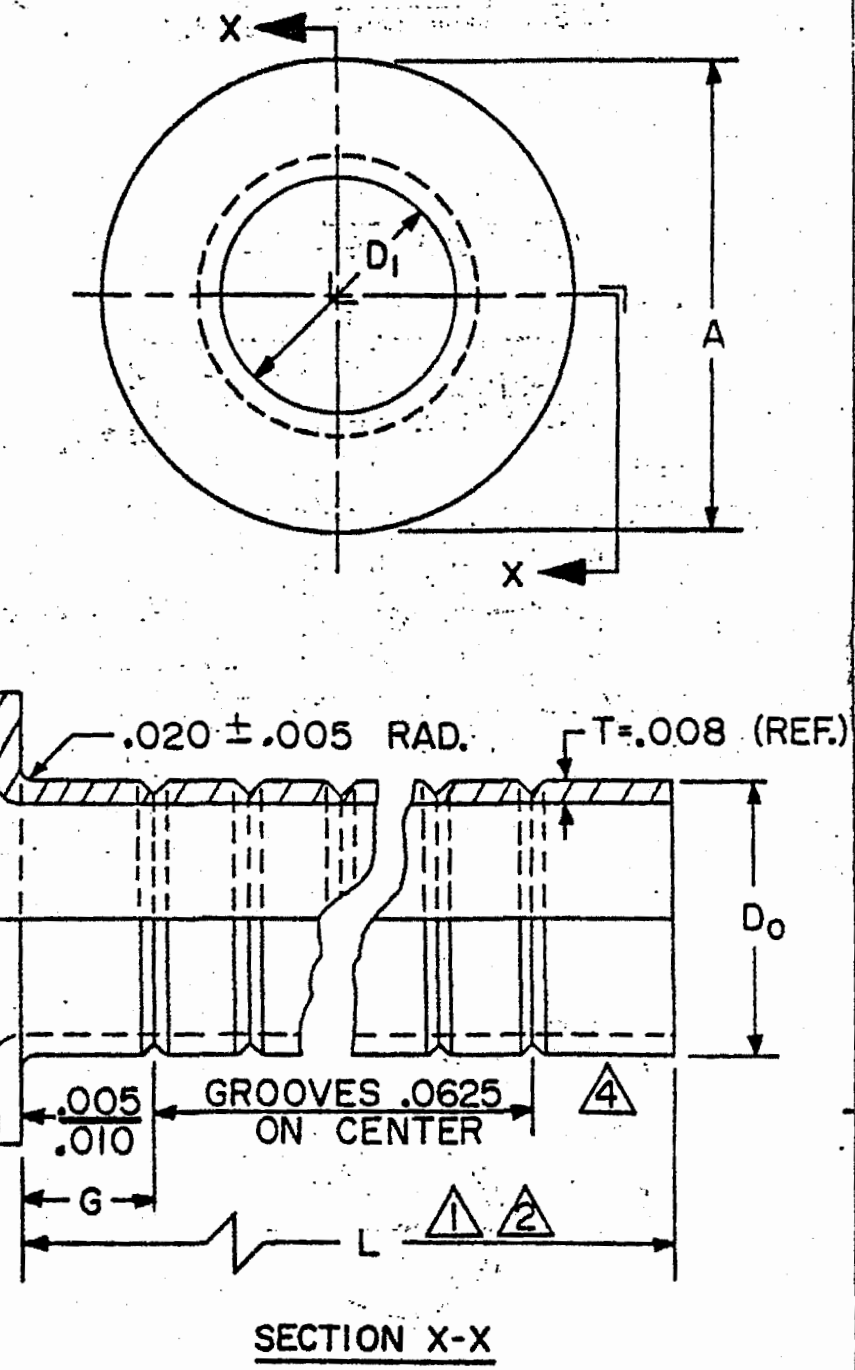


FIGURE 5. Protruding head, grooved.

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NOTES

1 Dimensions are in inches.

2 Sleeves supplied in standard lengths - adjust to required length when needed.

3 Recommend for metal-to-metal to clearance fit of sleeve to hole.

4 1/64 Oversize
Sleeves 06-08 ungrooved last 1/4
Sleeves 10-12 ungrooved last 5/16

1/32 Oversize
Sleeves 06-08 ungrooved last 1/4
Sleeves 10-12 ungrooved last 5/16

5 Tool and part numbers referenced herein are available from: J. O. King, Incorporated.

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